

File and Directory Handling Commands in Bash (Terminal Style)

Definition:

File and directory handling commands in Bash are used to create, view, modify, move, copy, delete, and manage files and folders in Linux systems.

1. File Handling Commands

cat	- Display contents of a file
tac	- Display file in reverse order
nl	- Display file with line numbers
head	- Show first lines of file
tail	- Show last lines of file
touch	- Create new empty file
cp	- Copy a file
mv	- Move or rename a file
rm	- Delete file

Examples:

```
cat notes.txt
nl notes.txt
head -5 notes.txt
tail -n 3 notes.txt
```

2. Directory Handling Commands

pwd	- Show current working directory
ls	- List directory contents
cd	- Change directory
mkdir	- Create a new directory
rmdir	- Remove empty directory
rm -r	- Remove directory and contents

cp -r - Copy directory recursively
mv - Move or rename a directory

Examples:

pwd
ls -lh
cd /etc
mkdir projects
rm -r old_folder

3. File Search and Filtering

find - Search for files or directories
locate - Quickly find files (uses database)
grep - Search for text patterns
wc - Count lines, words, or bytes
sort - Sort lines alphabetically
uniq - Remove duplicate lines
cut - Extract sections from each line

Examples:

find /etc -name '*.conf'
grep 'bash' /etc/passwd
wc -l file.txt
sort names.txt | uniq

4. File Compression and Archiving

tar - Archive files and directories
gzip - Compress files
gunzip - Decompress files
zip - Create zip archive
unzip - Extract zip archive

Examples:

```
tar -cvf backup.tar folder/
tar -xvf backup.tar
gzip notes.txt
gunzip notes.txt.gz
zip myzip file1 file2
unzip myzip.zip
```

5. File Permission and Ownership Commands

```
chmod    - Change file permissions
chown    - Change file owner
chgrp    - Change group ownership
umask    - Set default permission mask
```

Examples:

```
chmod +x script.sh
chmod 644 file.txt
chown user:user file.txt
chgrp developers file.txt
```

Summary Table

Category	Common Commands
File Ops	cat, nl, head, tail, cp, mv, rm, touch
Dir Ops	pwd, ls, cd, mkdir, rmdir, cp -r, rm -r
Search	find, locate, grep, wc, sort, uniq, cut
Archive	tar, gzip, zip, unzip, gunzip
Perms	chmod, chown, chgrp, umask